FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE – GRADE 10 | SUB-STRAND 1.4: TRANSPORT IN PLANTS	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation task. You have spent 8 lessons investigating how a maize plant moves water and nutrients from the soil to its topmost leaves — and what happens to that water in the end. Now it is time to bring everything you have learned together into one well-reasoned, evidence-based explanation.

Your goal is to fully answer the driving question:
"How does a maize plant on a Kenyan farm move water and nutrients from the soil all the way to its topmost leaves — and what happens to that water in the end?"

Read each section prompt carefully. Use evidence from the red-dye experiment you observed at the start of the unit (the maize seedling in Kisumu) and the polythene-bag transpiration demonstration. Connect your answer back to these two phenomena at every stage. Write in full sentences and use the scientific vocabulary you have built up across the 8 lessons. Each section builds on the one before it, so answer them in order.

Section 1: The Two Pathways — Xylem and Phloem

When the maize stem was cut crosswise after sitting in red dye, only certain rings turned red while the rest of the stem stayed white. Explain what this tells you about how the plant's transport system is organised. In your answer: (a) name and describe the two main transport tissues found in a maize stem; (b) explain what each tissue transports and in which direction; and (c) explain why only some parts of the cross-section turned red and others	The red-dye experiment reveals that a maize stem contains two separate transport tissues arranged in vascular bundles. The xylem is the tissue that turned red. It is made of hollow, dead cells joined end-to-end to form continuous tubes called vessels, and it transports water and dissolved mineral salts (such as nitrates, phosphates, and potassium absorbed from the Kenyan farm soil) upward — from the roots to the leaves. Because the red dye dissolved in the water, it travelled exclusively through the xylem, staining those rings red. The phloem is the tissue that stayed white. It is made of living sieve tube cells and companion cells, and it transports dissolved sugars (produced by photosynthesis in the leaves) both upward and downward to all parts of the plant that need energy — for example, to the growing maize cobs or to the roots. Because the phloem does not carry water the way xylem does, the dye never entered it, so it remained unstained. This separation of pathways means the plant can move raw materials and manufactured food simultaneously without the two systems interfering with each other.

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Section 2: How Water Enters the Plant — Osmosis and Root Hair Cells	
Before water can travel up the maize stem, it must first enter the root from the soil. Using your knowledge of osmosis and root hair cells, explain: (a) what osmosis is and why it is the process responsible for water entering the root; (b) the role of root hair cells in making this process efficient; and (c) what would happen to the maize plant on a farm in the Rift Valley if the farmer applied too much fertiliser to the soil around the roots.	Water enters the maize root by a process called osmosis — the movement of water molecules from a region of higher water concentration (the soil water) to a region of lower water concentration (the cell sap inside the root hair cells), through a partially permeable membrane. The root hair cells are the key structures that make this efficient: they are long, thin extensions of the outer root cells that greatly increase the surface area in contact with soil water. This large surface area means that more water molecules can cross the membrane at the same time, speeding up uptake. The cell membrane of each root hair cell acts as the partially permeable barrier — it allows water molecules to pass through but restricts larger dissolved molecules. Once water enters the root hair cells, it continues to move by osmosis from cell to cell across the cortex and eventually into the xylem vessels, which carry it upward. If a farmer in the Rift Valley applied too much fertiliser, the concentration of dissolved salts in the soil water would increase dramatically — possibly exceeding the concentration of salts inside the root hair cells. This would reverse the osmotic gradient: water would actually leave the root cells and move out into the soil by osmosis. The plant would wilt and could die, even if the soil appears wet. This condition is sometimes called fertiliser burn.

Section 3: The Journey Upward — Forces That Drive Water Through the Xylem	
Water must travel from the roots of a maize plant all the way up to its topmost leaves — sometimes over a metre against the force of gravity. Explain the three main forces (transpiration pull/cohesion-tension, root pressure, and capillary action) that work together to make this possible. For each force, describe what causes it and where in the plant it acts. Then explain how these forces combined can account for water reaching the very top of a tall maize plant on a farm in Kericho.	Three forces cooperate to pull and push water up the xylem of a maize plant. The first and most powerful is transpiration pull, which works together with the cohesion and tension properties of water. Water constantly evaporates from the leaf cells into the air spaces inside the leaf and then exits through tiny pores called stomata. This evaporation creates a water deficit in the leaf cells, which then draw water from the xylem vessels by osmosis. Because water molecules are strongly attracted to each other (cohesion), they form an unbroken column in the xylem — when water is pulled from the top, the entire column is pulled upward like a chain, creating tension throughout the vessel. This cohesion-tension mechanism is the main engine driving water to the top of the plant. The second force is root pressure. Mineral salts are actively pumped into the xylem in the roots, lowering the water concentration there. Water follows by osmosis, building up a positive pressure at the base of the xylem that pushes the water column upward. Root pressure alone is not strong enough to push water to the top of a tall maize plant, but it contributes, especially at night when transpiration is low. The third force is capillary action: the attraction between water molecules and the inner walls of the narrow xylem vessels (adhesion) helps water 'cling' to the walls and resist falling back down, supporting the water column. Together, these three forces form a continuous relay — root pressure initiates the push from below, capillary action supports the column, and transpiration pull provides the dominant pull from above — allowing water to reach the topmost leaf of a maize plant in Kericho even on a hot afternoon.

Section 4: Transpiration — Where Does the Water Go, and Why Does the Plant 'Lose' It?	
The polythene-bag	The water droplets inside the polythene bag are evidence of transpiration — the process by which a plant loses water vapour to the

demonstration showed water droplets forming inside the bag sealed around a leafy plant cutting left in afternoon sun — even though no liquid water was added to the bag. Using this observation as your evidence: (a) define transpiration and explain exactly where in the leaf the water vapour comes from; (b) describe the role of stomata and guard cells in controlling this water loss; and (c) explain why a plant continues to transpire even though losing water puts it at risk of wilting — what essential benefit does transpiration provide to the plant?	atmosphere. Here is how it happens: water absorbed by the roots travels up the xylem and enters the leaf mesophyll cells. These cells are surrounded by moist air spaces inside the leaf. Water evaporates from the surface of the mesophyll cells into these air spaces, and the water vapour then diffuses out of the leaf through small pores called stomata, moving from the high concentration inside the leaf to the lower concentration in the outside air. Because the polythene bag traps the escaping water vapour, it condenses on the cooler plastic surface and appears as visible droplets. The stomata are controlled by pairs of guard cells. When the plant has plenty of water and light is available, the guard cells absorb water by osmosis, become turgid, and bow outward — opening the stomatal pore. When the plant is water-stressed or it is dark, the guard cells lose water, become flaccid, and the pore closes, reducing water loss. Even so, a healthy maize plant on a sunny Kenyan afternoon keeps its stomata open during daylight because transpiration provides two vital benefits: first, it is the engine that drives the transpiration pull, which is the primary force moving water and dissolved minerals from the roots all the way to the top of the plant — without transpiration, nutrient delivery would largely stop; second, the evaporation of water from leaf cells has a cooling effect, preventing the leaf from overheating in the strong equatorial sun. So transpiration is not simply 'waste' — it is a necessary cost the plant pays to keep nutrients flowing and its temperature regulated.

Section 5: Bringing It All Together — Answering the Driving Question	
Now write a complete, connected answer to the driving question: 'How does a maize plant on a Kenyan farm move water and nutrients from the soil all the way to its topmost leaves — and what happens to that water in the end?' Your answer must: (a) trace the full journey of a water molecule from the soil in Kisumu to the topmost leaf of a maize plant; (b) explain what drives each stage of the journey; (c) state what happens to the water at the end of the journey; and (d) reflect on why this system matters to the farmer — linking to food security or crop yield in Kenya.	A water molecule's journey in a maize plant begins in the soil on a Kenyan farm. The molecule moves by osmosis from the soil water, which has a relatively high water concentration, into the cytoplasm of a root hair cell, which has a lower water concentration because it contains dissolved mineral salts absorbed from the soil. The water molecule continues to move by osmosis from cell to cell across the root cortex until it reaches the xylem vessel at the centre of the root. Here it joins the continuous water column inside the hollow xylem tube. Three forces work together to push and pull it upward against gravity: root pressure (caused by active pumping of mineral salts into the xylem) pushes from below; capillary action (adhesion between the water molecule and the xylem wall) prevents it from falling back; and — most importantly — transpiration pull drags the entire water column upward as water molecules evaporate from the leaf cells into the leaf's air spaces and escape through the open stomata. Because water molecules are cohesive, the molecule in the root is pulled upward as part of an unbroken chain all the way to the top leaf. Dissolved mineral nutrients — nitrates, phosphates, and potassium — travel with the water molecule in this same xylem stream, supplying the leaf cells with the raw materials needed for growth and photosynthesis. Once the water molecule reaches the mesophyll cells of the topmost leaf, most of it evaporates from the cell walls into the internal air spaces and diffuses out through the stomata as water vapour — the process called transpiration. A smaller portion is used directly in the light-dependent reactions of photosynthesis, where it is split to release oxygen and hydrogen. The sugars made by photosynthesis are then loaded into the phloem and transported in all directions to growing, respiring, or storing parts of the maize plant. This system matters enormously to Kenyan farmers: if the xylem is blocked (for example, by a fungal wilt disease), or if the roots are damaged by excess fertiliser or drought, the water and nutrient supply to the leaves fails — photosynthesis slows, the plant wilts, and the maize cobs do not fill properly, directly reducing the harvest that feeds families and supplies markets across Kenya.

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Scientific Accuracy of Plant Transport Concepts	All explanations of xylem, phloem, osmosis, transpiration pull, root pressure, capillary action, stomata, and guard cells are scientifically correct and precisely worded. No misconceptions present.	Most concepts are correctly explained with minor imprecision (e.g., direction of transport slightly unclear or one force poorly described). No major misconceptions.	Several key concepts are incorrect or confused (e.g., xylem and phloem roles swapped, osmosis described as diffusion of all molecules, or transpiration pull missing). Major misconceptions evident.
Use of Evidence from the Phenomena	Both the red-dye maize experiment and the polythene-bag transpiration demonstration are explicitly referenced as evidence in relevant sections. Observations are accurately described and directly linked to the scientific explanation.	At least one phenomenon is referenced with accurate observation, but the link to the scientific explanation is partially developed or one phenomenon is omitted.	Neither phenomenon is meaningfully used as evidence, or observations are described inaccurately. The explanation reads as abstract recall rather than evidence-based reasoning.
Completeness of the Water Journey (Driving Question Answer)	Section 5 traces the complete, unbroken journey of a water molecule from soil → root hair cell → xylem → leaf mesophyll → stomata → atmosphere (transpiration) or photosynthesis, with the driving force at each stage correctly identified.	The journey is mostly complete but one stage or one driving force is missing or vaguely described (e.g., the xylem-to-leaf transition is skipped, or transpiration pull is named but not explained).	The journey is incomplete, covering fewer than three stages, or the driving forces are absent or incorrect throughout.
Use of Scientific Vocabulary	All key terms (xylem, phloem, osmosis, transpiration, stomata, guard cells, cohesion, tension, root pressure, capillary action, turgid, flaccid, vascular bundle, translocation) are used correctly and consistently throughout all sections.	Most key terms are used correctly. One or two terms are missing or used slightly out of context, but the overall meaning is clear.	Few key terms are used, or several are used incorrectly (e.g., 'osmosis' used to mean any movement, 'transpiration' confused with respiration). The explanation relies mainly on everyday language.
Real-World Connection to Kenyan Farming Context	The explanation makes a clear, specific, and accurate connection between the plant transport system and its significance for Kenyan farmers — e.g., consequences of drought, wilt disease, over-fertilisation, or crop yield — demonstrating understanding of why this biology matters in a local food security context.	A connection to farming or food security in Kenya is made but is general or briefly stated (e.g., 'if transport fails, the plant dies and the farmer loses crops') without elaborating on a specific mechanism or condition.	No connection to Kenyan farming or food security is made, or the connection is factually incorrect.